

Problem Statement: AgriSense AI

Domain: Agriculture | Model: IBM Granite (ibm/granite-4-h-small)

The Challenge

Smallholder agriculture remains acutely vulnerable to rapid weather shifts, sudden pathogen outbreaks, and uncalibrated chemical use:

- **Epiphytotic Vulnerability:** Fungal pathogens like Late Blight and Anthracnose cause severe crop loss when humidity and rain cross biological thresholds before manual detection.
- **Agrochemical Overdosing:** Cultivators apply excessive broad-spectrum chemicals without diagnostic guidance, causing phytotoxicity and soil degradation.
- **LLM Hallucination Hazards:** General-purpose AI tools regularly recommend inaccurate dosages and dangerous tank mixtures not grounded in verified packages of practices.
- **Vernacular Language Barriers:** Critical agrometeorological advisories rarely reach farmers in regional native dialects.

The Objective

Build AgriSense AI, an autonomous agronomic decision platform powered by IBM watsonx.ai and IBM Granite:

- Autonomous Sentinel Daemon: Continuously monitor hyper-local weather telemetry (temperature, humidity, rain probability) to issue zero-latency risk alerts.
- Multi-Modal Pathology Vision: Segment foliar lesions and fruit rot craters from field photos to classify plant infections.
- ICAR-Grounded RAG Retrieval: Fetch verified active ingredients and exact per-liter dilution formulas from agronomic research datasets.
- Granite Decision Fusion: Deploy Granite-4-H-Small to generate actionable 7-day field schedules and simulated chemical trade-offs in structured JSON.
- Vernacular Voice Delivery: Deliver complete operational decisions in Kannada, Telugu, Hindi, and English with audio speech synthesis.

Projected Field Outcomes

- Yield Loss Prevention: Preempt active fungal sporulation with a 48-to-72 hour lead time.
- Input Cost Optimization: Reduce excess chemical use by timing spraying strictly to weather-pathology infection windows.
- Zero-Hallucination Safety: Ground every curative spray recommendation strictly in vetted ICAR university protocols.